Prospective evaluation of a clinical decision support system in psychological therapy.

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CLINICAL & AGENTIC AI TAKEAWAY

A clinical decision support system in psychological therapy can enhance patient outcomes when successfully implemented and incorporated into clinical practice.

ABSTRACT

Objective: Thus far, most applications in precision mental health have not been evaluated prospectively. This article presents the results of a prospective randomized-controlled trial investigating the effects of a digital decision support and feedback system, which includes two components of patient-specific recommendations: (a) a clinical strategy recommendation and (b) adaptive recommendations for patients at risk for treatment failure. Method: Therapist-patient dyads (N = 538) in a cognitive behavioral therapy outpatient clinic were randomized to either having access to a decision support system (intervention group; n = 335) or not (treatment as usual; n = 203). First, treatment strategy recommendations (problem-solving, motivation-oriented, or a mix of both strategies) for the first 10 sessions were evaluated. Second, the effect of psychometric feedback enhanced with clinical problem-solving tools on treatment outcome was investigated. Results: The prospective evaluation showed a differential effect size of about 0.3 when therapists followed the recommended treatment strategy in the first 10 sessions. Moreover, the linear mixed models revealed therapist symptom awareness and therapist attitude and confidence as significant predictors of an outcome as well as therapist-rated usefulness of feedback as a significant moderator of the feedback-outcome and the not on track-outcome associations. However, no main effects were found for feedback. Conclusions: The results demonstrate the importance of prospective studies and the high-quality implementation of digital decision support tools in clinical practice. Therapists seem to be able to learn from such systems and incorporate them into their clinical practice to enhance patient outcomes, but only when implementation is successful. (PsycInfo Database Record (c) 2022 APA, all rights reserved).

Study Overview & Background

Objective: Thus far, most applications in precision mental health have not been evaluated prospectively. This article presents the results of a prospective randomized-controlled trial investigating the effects of a digital decision support and feedback system, which includes two components of patient-specific recommendations: (a) a clinical strategy recommendation and (b) adaptive recommendations for patients at risk for treatment failure. Method: Therapist-patient dyads (N = 538) in a cognitive behavioral therapy outpatient clinic were randomized to either having access to a decision

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Methodology & Clinical Focus

Study Classification: rct. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

A clinical decision support system in psychological therapy can enhance patient outcomes when successfully implemented and incorporated into clinical practice.

Extended Repository Abstract

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Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

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